Writing Quadratic Functions

Vertex?

Standard/General Form: $f(x) = ax^2 + bx + c \quad a \neq 0$ ________________

Vertex Form: $f(x) = a(x-h)^2 + k \quad a \neq 0$ ________________

Intercept/Factored Form: $f(x) = a(x-p)(x-q) \quad a \neq 0$ ________________

Example 1: Write the following quadratic equations in both vertex form and intercept form. You will need to use your graphing calculator to determine the x-intercepts.

A) $f(x) = x^2 - 8x + 12$

B) $f(x) = -x^2 - 4x$

C) $f(x) = -\frac{1}{2}x^2 + \frac{1}{2}x + 3$

Example 2: Write the quadratic equation in vertex form given the vertex and one point on the parabola.

A) Vertex $(3,1)$, point $(1,9)$

B) Vertex $(-1,6)$, point $(-3,4)$

Example 3: Write the quadratic equation in factored form given the following information:

A) zeros $(-4,0),(0,0)$, point $(-3,6)$

B) zeros $(-3,0),(1,0)$, point $(2,4)$